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Multi-Agent Simulation of AI-Mediated Prior Authorization for Specialty Medications

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Abstract

Insurers and providers are both deploying AI in prior authorization, raising concerns of a prisoner's dilemma where mutual adoption produces collectively harmful outcomes. We develop a multi-agent simulation where LLM-driven provider and insurer agents negotiate authorization for a specialty medication under information asymmetry, with cooperative, adversarial, or no-guidance strategies injected via prompts. In a 3×3 factorial experiment on a single gray-zone clinical case, we find that the insurer's choice between requesting additional information and outright denial at initial review is the mechanism through which gray-zone ambiguity is resolved. Default (no-guidance) insurer agents denied the drug while approving diagnostics, applying policy literally rather than interpretively, suggesting that AI adjudication systems deployed without explicit guidance may default to denial in ambiguous cases. Game-theoretic analysis shows the game's classification is parameter-dependent: at observed administrative costs ($5.79/turn for providers vs. $0.05/turn for insurers, a 116× asymmetry), the game is an asymmetric exploitation where denial is nearly costless for insurers; under richer cost models incorporating litigation, reputational, and network costs, it can transition to the prisoner's dilemma hypothesized by prior work.

1. Introduction

Prior authorization (PA) is a utilization management process requiring providers to obtain insurer approval before rendering certain services. Insurers adjudicate requests by approving, modifying, denying, or requesting documentation; providers may appeal through successive review levels. This creates substantial administrative burden: physicians complete 39 PA requests weekly, spending 13 hours on PA-related tasks (American Medical Association, 2024). Both sides are now deploying artificial intelligence to gain advantage: insurers use AI to scrutinize claims and increase denials, while providers deploy AI to generate appeals and optimize documentation (Williams, 2024).

Moritz et al. (2025) argue that this dynamic resembles a prisoner's dilemma, warning that "each party's rational decision to adopt AI for self-protection creates collectively harmful outcomes for everyone." The incentive structure is clear: if one party deploys AI, the other must respond or face disadvantage. Yet mutual AI deployment may produce worse outcomes than mutual restraint: more administrative burden, more denials, more appeals, with no improvement in patient care. This raises a fundamental question: what happens when both provider and insurer use large language models to interpret ambiguous coverage policy? Does the predicted coordination failure actually emerge, and through what mechanism?

We address this through a controlled simulation study. Our contributions are:

1. **Multi-agent PA simulation framework.** We develop an environment where LLM-driven provider and insurer agents negotiate prior authorization through iterative submission, adjudication, and appeal cycles. Agents operate under information asymmetry (clinical guidelines vs. coverage policy) with behavior controlled via prompt-based strategy injection (cooperate, defect, or no guidance).

2. **3×3 factorial experiment.** On a single gray-zone clinical case, we cross three

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provider strategies with three insurer strategies and identify a `pending_info` vs. denial mechanism through which gray-zone ambiguity is resolved. Default (no-guidance) insurer agents apply policy literally, denying gray-zone treatments without cooperative interpretation.

3. **Game-theoretic payoff analysis.** We derive utility functions from simulation outcomes and show the game's classification is parameter-dependent: at observed administrative costs, a 116× cost asymmetry makes denial nearly costless for insurers, producing an exploitation structure; under richer cost models, the game can transition to the prisoner's dilemma hypothesized.

2. Related Work

AI and machine learning are being deployed on both sides of healthcare administration. On the insurer side, a scoping review identifies eight application domains across health insurance, including claims processing, fraud detection, utilization monitoring, and policy decision support (Ramezani et al., 2025). On the provider side, Owolabi (2025) show that elastic net classifiers can predict which hospital admission denials will overturn on appeal with 84% accuracy; post-deployment, monthly appeal volumes and overturned denials nearly doubled. These systems each optimize for one party independently; neither models the interaction dynamics that emerge when both sides deploy AI simultaneously.

Multi-agent LLM simulation offers a methodology for studying such interactions. Fan et al. (2025) introduce AI Hospital, where doctor, patient, examiner, and chief physician agents interact across multi-turn diagnostic conversations. Their work demonstrates that LLM diagnostic performance degrades substantially in interactive settings relative to static benchmarks (even GPT-4 achieves less than half the upper-bound score) and that a multi-doctor dispute resolution mechanism can partially recover accuracy. This establishes that multi-agent simulation surfaces emergent behaviors not visible in single-agent evaluation.

Our work connects these threads. The single-agent systems above optimize for one side of the PA transaction; we model both sides simultaneously within a shared encounter, making the interaction itself the object of study. Unlike collaborative multi-agent settings where agents share a diagnostic goal, our agents hold opposing objectives (the provider maximizes approved services while the insurer minimizes unnecessary coverage) and operate under information asymmetry, each receiving a different document (clinical guideline vs. coverage policy). We further introduce strategy as an experimental variable via prompt injection, enabling controlled comparison of cooperative and adversarial postures within the same clinical scenario.

3. Methods

3.1. Case and Policy Selection

Demonstrating AI coordination failure in prior authorization requires: (1) complete patient data including demographics, history, labs, and medications; (2) a treatment context where PA is required and known to create friction; and (3) a genuine policy gray zone where strategy can determine interpretation.

We selected infliximab for Crohn's disease from a published case report (Lv et al., 2021): a 60-year-old female with Crohn's disease, small intestinal stricturing, partial response to mesalazine 3 g/day, elevated fecal calprotectin (827 μg/g, indicating active inflammation), and cytopenias (WBC 3.2×10^9/L, platelets 123×10^9/L). Full patient data are in Appendix A.

Each agent receives a different document, creating information asymmetry that mirrors real-world practice. The provider agent receives the AGA 2021 clinical guideline (Feuerstein et al., 2021), which suggests early biologic introduction for moderate-to-severe Crohn's disease without requiring prior steroid failure. The insurer agent receives Cigna coverage policy IP0660 (Cigna, 2026), a checklist-style step-therapy document that requires documentation of steroid trial or contraindication, conventional systemic therapy trial, fistulas, or ileocolonic resection, and explicitly excludes mesalamine as qualifying systemic therapy. Full document texts are in Appendix B.

These documents point in opposite directions for this patient: the clinical guideline supports immediate infliximab, while the coverage policy demands a step-therapy prerequisite that the patient has not formally met.

The interaction between these conflicting documents creates a genuine interpretive gray zone centered on a single pivot point. Three of the four qualifying pathways are clearly unmet: no conventional immunomodulator trial (the patient's cytopenias make azathioprine/6-MP risky), no fistulizing disease, and no prior resection. Everything therefore hinges on the remaining steroid contraindication. The patient's cytopenias make corticosteroids clinically risky (steroids can further suppress bone marrow), but this risk is not a formal, documented contraindication. Whether cytopenias-as-steroid-risk satisfies the coverage policy's contraindication criterion depends on interpretive stance: a good-faith reading says yes (steroids were considered but are unsafe); a literal reading says no (no formal contraindication is documented). This case thus serves as an existence proof: the clinical facts create a genuine ambiguity that the experiment tests agents' ability to resolve (Section 5.1).

3.2. Simulation Design

The simulation proceeds in four phases (Figure 1). Phase 1 presents the patient case to the agents (currently deterministic; see Appendix H for the noise-injection design). Phase 2 (Utilization Review) and Phase 3 (Claims Adjudication) contain the strategic interaction. Phase 4 computes the financial settlement (currently per-line reimbursement; see Appendix H for the planned extensions).

In Phase 2, the provider and insurer negotiate over prior authorization through alternating submissions and adjudications. The provider requests services with clinical justification; the insurer evaluates each request against coverage policy. When the insurer approves a lab or imaging order, the Environment returns results from the patient record (or, if no result exists in the source data, generates a clinically appropriate one), allowing the provider to use new findings in subsequent requests. Phase 2 continues until every requested service reaches a terminal state. Phase 3 follows the same logic but only for services that were approved or that the provider chose to deliver despite denial.

We model three review levels grounded in the Medicare appeal structure (Centers for Medicare & Medicaid Services, 2024): initial determination (Level 0), first appeal (Level 1), and independent external review (Level 2). The simulation terminates at Level 2 because Levels 0–2 encompass the purely administrative interaction loop: the domain where AI automation is currently being deployed. Further legal appeals (ALJ, federal court) involve formal proceedings with different rules of evidence, placing them outside AI-mediated negotiation. Appendix F maps these to Medicare's five-level system.

3.3. Strategy Operationalization

We operationalize three strategy conditions. In the **default** condition, no strategy guidance is injected, revealing the LLM's baseline disposition in gray-zone situations. In the **cooperate** and **defect** conditions, we inject interpretive guidance into the system prompt. All conditions share the same clinical facts, clinical guideline, and coverage policy; they differ only in ambiguity-resolution framing:

Provider cooperate: "Choose the minimum clinically defensible set of requested services and the lowest reasonable intensity that still supports timely, appropriate treatment."

Provider defect: "Choose the maximum plausibly defensible set of requested services and the highest plausible intensity supported by the case facts/policy. Include additional supportive line items if remotely justifiable; do not fabricate facts."

Insurer cooperate: "Apply policy in a good-faith, access-preserving way. If the request plausibly satisfies policy, approve. Prefer approve over deny when policy interpretation is genuinely ambiguous."

Insurer defect: "Apply policy in a strict, cost-containment way. If the request is not clearly supported by explicit policy criteria and documentation, deny or modify/downgrade. Prefer deny over approve when evidence is borderline."

Full prompts are provided in Appendix C.

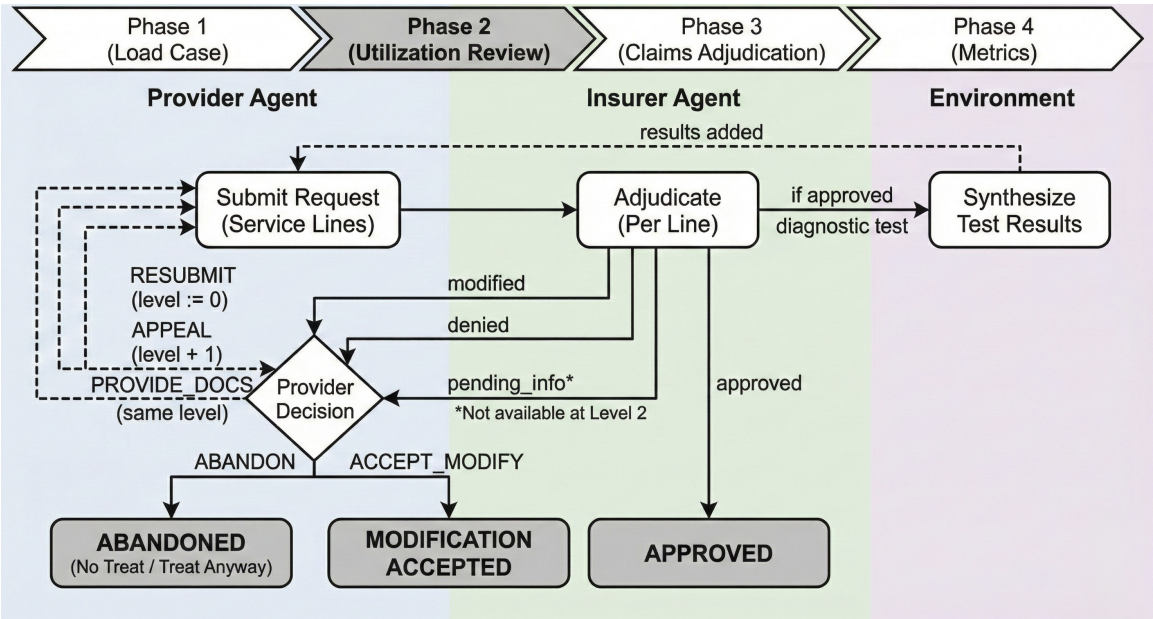

Figure 1: Phase 2 (Utilization Review) simulation flow. Phase 3 (Claims Adjudication) follows the same structure for delivered services, with WRITE_OFF replacing the abandon modes. Full agent specifications are in Appendix D.

3.4. Experimental Design

We employ a 3×3 factorial design crossing provider strategy (cooperate, defect, default) with insurer strategy (cooperate, defect, default), yielding the nine conditions shown in Table 1.

4. Results

Table 2 summarizes primary outcomes across all nine experimental conditions.

4.1. Condition Summaries

Across the original four conditions, the cooperate/defect strategy injections produced expected behavior: cooperating-insurer conditions ended in full approval, defecting-insurer conditions in complete denial, validating that the simulation responds to strategy manipulation. The five default-strategy conditions reveal a more nuanced picture.

CP_CI. The provider requested 2 lines: infliximab (J1745, 3 induction infusions) and an acute viral hepatitis panel (80074). The insurer issued `pending_info` requesting step-therapy documentation and biosimilar confirmation. The provider switched to biosimilar infliximab-axxq (Q5121), cited cytopenias (WBC 3.2×10^9/L, platelets 123×10^9/L) as steroid contraindication, and confirmed GI prescriber involvement. The insurer approved both lines. The Environment synthesized hepatitis panel results (1 synthesis event).

CP_DI. The provider requested 4 lines: infliximab (J1745) plus individual hepatitis B serologies (HBsAg, anti-HBc, anti-HBs). The insurer denied all lines without `pending_info`, citing unmet step-therapy. The provider resubmitted three times with the same cytopenias argument; all were rejected. By turn 3, the provider simplified to infliximab alone, which was still denied. The provider chose `ABANDON` with `NO_TREAT`. No diagnostic tests were approved, so no Environment synthesis occurred.

CP_NI. The provider requested 5 lines: infliximab-dyyb biosimilar (Q5102, 3 induction infusions), IV administration (96413), and hepatitis B serologies (87340, 86704, 86706). The insurer denied the drug and IV administration lines without `pending_info`, citing unmet step-therapy, but approved all 3 screening labs. The provider chose `ABANDON` with `NO_TREAT` for the de-

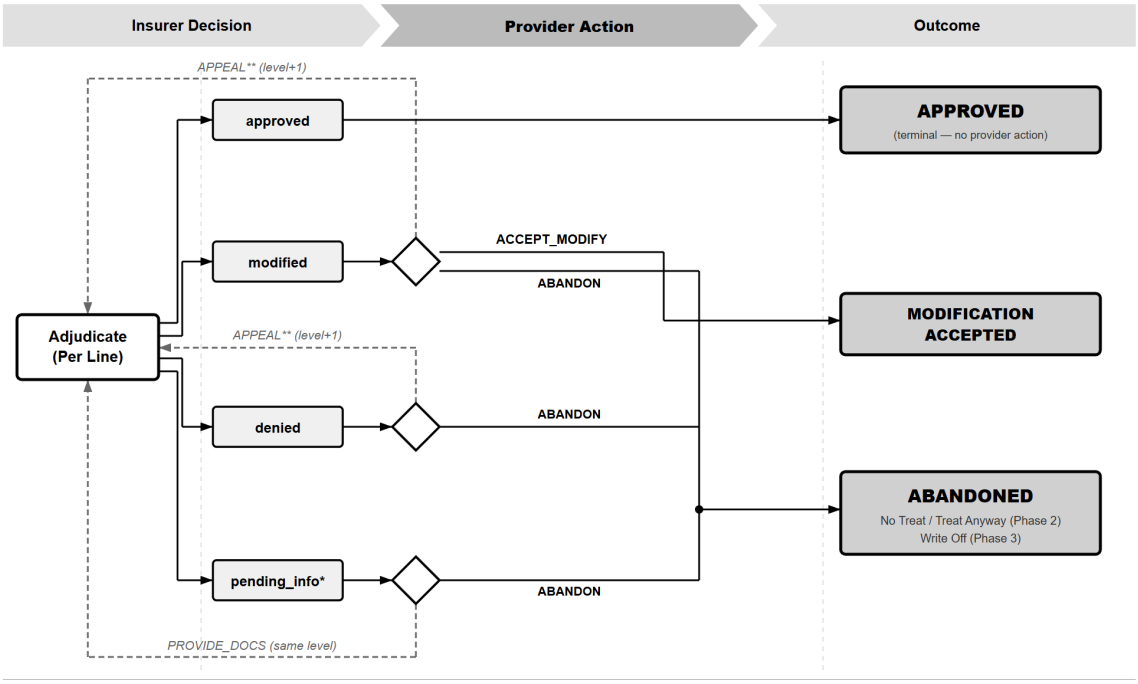

Figure 2: Provider per-line action space, expanding the provider decision node in Figure 1. At any point the provider may also RESUBMIT (bundle-level), withdrawing all lines and restarting at Level 0 (see Figure 1). * pending_info is not issued by the insurer at Level 2. ** APPEAL is not available at Level 2 (maximum review level).

Table 1: Experimental conditions. C = Cooperate, D = Defect, N = Default (no guidance).

	Insurer C	**Insurer D**	**Insurer N**
Provider C	CP_CI	CP_DI	CP_NI
Provider D	DP_CI	DP_DI	DP_NI
Provider N	NP_CI	NP_DI	NP_NI

nied treatment lines. Phase 2 resolved in 1 turn with no appeals. Three diagnostic lines were delivered and paid.

DP_CI. The provider requested 11 lines: infliximab (J1745, 5 infusions), IV administration (96365), hepatitis B panel (3 tests), CMP (80053), CBC (85025), CRP (86140), MRI abdomen (74183), MRI pelvis (72197), and fecal calprotectin (83993). The insurer issued `pending_info` for treatment lines while immediately approving all 9 diagnostic tests, triggering 9 Environment synthesis events (serologies, metabolic panel, CBC, CRP, MRI, fecal calprotectin). After provider documentation, the insurer approved all 11 lines, switching the drug to biosimilar infliximab-abda under code Q5105.[1]

DP_DI. The provider requested 7 lines: infliximab-axxq biosimilar (Q5121, 3 induction infusions), IV administration (96413), and screening labs (HBsAg, anti-HBc, anti-HBs, HCV Ab, CRP). The insurer denied all 7 lines without `pending_info`. The provider resubmitted, then escalated to Level 1 appeal (Medical Director), then Level 2 (IRE). All denied at every level. The provider chose `ABANDON` with

1. Q5105 is actually Retacrit (epoetin alfa-epbx), not an infliximab product. The insurer agent hallucinated this code-drug mapping. The correct code for infliximab-abda (Renflexis) is Q5104; we use Q5104 pricing in the payoff analysis (Section 4.4).

Table 2: Primary outcomes by strategy condition. P2 Turns = Phase 2 submission-response cycles. Lines Req = total unique service lines requested. Delivered = services rendered to patient. Paid = claims reimbursed in Phase 3.

Condition	Provider	Insurer	P2 Turns	Lines Req	Delivered	Paid
CP_CI	Cooperate	Cooperate	2	2	2	2
CP_DI	Cooperate	Defect	4	4	0	0
CP_NI	Cooperate	Default	1	5	3	3
DP_CI	Defect	Cooperate	2	11	11	11
DP_DI	Defect	Defect	4	7	0	0
DP_NI	Defect	Default	3	13	9	9
NP_CI	Default	Cooperate	2	5	5	5
NP_DI	Default	Defect	3	4	1	1
NP_NI	Default	Default	6	2	0	0

`NO_TREAT`. As in CP_DI, no Environment synthesis occurred.

DP_NI. The provider requested 13 lines: infliximab-dyyb biosimilar (Q5102, 9 infusions), IV administration (96413, 96415 additional hour), diphenhydramine premedication (J1200), hepatitis B serologies (87340, 86704, 86706), CBC (85025), CMP (80053), CRP (86140), MRI abdomen (74183), MRI pelvis (72197), and fecal calprotectin (83993). The insurer denied all treatment lines (drug, IV, premedication) without `pending_info`, citing unmet step-therapy, but approved all 9 diagnostic lines. The provider appealed through Level 1 and Level 2; treatment lines remained denied at every level. The provider chose `ABANDON` with `NO_TREAT` for the 4 denied lines. Nine diagnostic lines were delivered and paid.

NP_CI. The provider requested 5 lines: infliximab (J1745, 3 induction infusions), IV administration (96413), MRI abdomen (74183), MRI pelvis (72197), and fecal calprotectin (83993). The insurer issued `pending_info` requesting step-therapy documentation for the drug, modified the MRI abdomen line, and approved the remaining diagnostic tests. The provider accepted the modification and supplied documentation. The insurer approved all lines. All 5 lines were delivered and paid.

NP_DI. The provider requested 4 lines: infliximab-axxq biosimilar (Q5121, 3 induction infusions), IV administration (96413), MRI abdomen (74183), and MRI pelvis (72197). The insurer denied all 4 lines without `pending_info`. The provider appealed through Level 1 (Medical Director) and Level 2 (IRE). At Level 2, the reviewer approved infliximab but upheld denial of the remaining 3 lines. The provider chose `ABANDON` with `NO_TREAT` for the denied lines. One line was delivered and paid.

NP_NI. The provider requested 2 lines: infliximab (J1745, 9 infusions) and IV administration (96413, 9 visits). The insurer denied both lines without `pending_info`, citing unmet step-therapy. The provider resubmitted with reduced quantities, then appealed through all levels (4 appeals over 6 turns). All denied at every level. The provider chose `ABANDON` with `NO_TREAT`. No services were delivered; this was the most protracted negotiation across all nine conditions.

4.2. Service Line Composition

Table 3 shows service lines requested by category.

Cooperate providers requested 2–5 lines; defect providers 7–13; default providers 2–5. Defect providers consistently requested the most ancillary services (IV administration, premedication, baseline labs, imaging, and monitoring) while cooperate and default providers requested narrower service sets.

4.3. Default Agent Behavior

Default insurer. The default insurer denied the drug (infliximab) in all three conditions (CP_NI, DP_NI, NP_NI) while approving diagnostic tests when requested (CP_NI: 3/3 labs; DP_NI: 9/9 diagnostics). This contrasts with

Table 3: Service lines requested by category and condition.

Category	CP_CI	CP_DI	CP_NI	DP_CI	DP_DI	DP_NI	NP_CI	NP_DI	NP_NI
Treatment (drug)	1	1	1	1	1	2	1	1	1
IV administration	0	0	1	1	1	2	1	1	1
Hepatitis screening	1	3	3	3	4	3	0	0	0
Baseline labs	0	0	0	3	1	3	0	0	0
Imaging	0	0	0	2	0	2	2	2	0
Monitoring	0	0	0	1	0	1	1	0	0
Total	**2**	**4**	**5**	**11**	**7**	**13**	**5**	**4**	**2**

both the cooperating insurer (which approved everything after `pending_info`) and the defecting insurer (which denied everything including diagnostics). The default insurer never used `pending_info`, matching the defecting insurer's posture for treatment lines, but diverged by approving ancillary services that did not trigger step-therapy requirements. This "partial defection" aligns with how the policy is structured: Cigna IP0660 imposes step-therapy requirements only on the biologic drug, not on diagnostic tests. Without interpretive guidance, the default insurer applied policy literally, denying where explicit criteria were unmet and approving where no restrictive criteria existed.

Default provider. Default providers requested 2–5 lines, comparable in scope to cooperate providers, but with a distinct composition. All three default-provider conditions included IV administration (CPT 96413), which cooperate providers in CP_CI and CP_DI omitted. Two of three (NP_CI, NP_DI) included imaging (MRI abdomen and pelvis), but none included hepatitis screening or baseline labs (Table 3). The default provider's appeal behavior was insurer-dependent: when the cooperating insurer issued `pending_info` (NP_CI), the provider supplied documentation, accepted a modification, and resolved in 2 turns with no appeals. When faced with denial (NP_DI, NP_NI), the provider appealed through the maximum review level before abandoning. NP_NI was the most protracted condition in the experiment (6 turns, 4 appeals), with the provider modifying its request mid-negotiation by reducing infusion quantity from 9 to 3 and changing the diagnosis code to emphasize the obstructive phenotype.

Interaction at NP_NI. When both agents receive no guidance, the outcome collapsed to complete denial: the default provider requested only 2 lines (drug + IV admin), both subject to step-therapy, and the default insurer denied both. This matches the defect-defect outcome but through different mechanisms: a narrow request scope rather than blanket denial, and persistent appeals rather than immediate abandonment.

4.4. Utility Function and Payoff Analysis

To formalize the strategic structure, we define agent utility functions over simulation outcomes. Let R^P denote the total reimbursement the provider receives for approved and paid services, let S^I denote the total value of services on the final submitted request (the insurer's exposure), and let $C_P = n \cdot c_P$ and $C_I = n \cdot c_I$ denote the total administrative costs for provider and insurer respectively, where n is the number of Phase 2 negotiation turns and c_P, c_I are per-turn costs. We define:

$$U_P = R^P - \alpha_P \cdot C_P \tag{1}$$

$$U_I = S^I - R^P - \alpha_I \cdot C_I \tag{2}$$

where $\alpha_P, \alpha_I \geq 0$ weight how much each agent values administrative burden relative to financial outcomes. The provider maximizes reimbursement net of admin burden. The insurer's utility reflects the value of denied services ($S^I - R^P$, which equals S^I when everything is denied and 0 when everything is approved) minus administrative burden. When $\alpha = 0$, admin costs are ignored; when $\alpha = 1$, a dollar of admin cost is weighted equally to a dollar of reimbursement or savings.

We price services using CMS reimbursement rates: drug costs from CMS ASP Q4 2025, lab costs from Clinical Lab Fee Schedule 2025, and per-turn administrative costs from the CAQH 2023 Index ($c_P = \$5.79$, $c_I = \$0.05$ for electronic PA). Four procedure codes (96365, 96413, 74183, 72197) use benchmark estimates bounded by verified CMS PFS facility rates; these account for less than 15% of total service value in any condition. Full pricing sources are in Appendix G.

We first analyze the 2×2 cooperate-defect subgame. Table 4 presents the payoff matrix at $\alpha_P = \alpha_I = 1$ (admin costs weighted at face value); we then examine how the game's structure changes under other parameterizations.

The insurer has a strictly dominant strategy to defect: $U_I^D > U_I^C$ regardless of provider strategy (\$2,828 > −\$0.10 when the provider cooperates; \$2,274 > −\$0.10 when the provider defects). Denying all services and retaining thousands of dollars in savings dominates paying out reimbursement, even after accounting for negligible admin costs. The provider has a *weakly* dominant strategy to defect: strictly preferred when the insurer cooperates (\$5,157 > \$1,868), tied when the insurer defects (−\$23 = −\$23). The Nash equilibrium is mutual defection (−\$23, \$2,274).

Whether this constitutes a prisoner's dilemma depends on parameterization. A prisoner's dilemma requires mutual cooperation to Pareto-dominate mutual defection for both players. At $\alpha = 1$ with observed per-turn costs, the insurer *prefers* mutual defection (\$2,274) to mutual cooperation (−\$0.10): denied-service savings dwarf the trivial admin cost. Under these parameters, the game is an asymmetric dominant-strategy exploitation rather than a PD. Both players defect at equilibrium, but for different reasons: the insurer defects because denial savings always exceed admin costs; the provider defects because, conditional on insurer cooperation, padding services extracts more reimbursement, and conditional on insurer defection, the outcome collapses to pure admin loss regardless of strategy. However, the PD structure emerges under parameterizations where insurer defection costs are higher, either through larger α_I (weighting administrative burden more heavily) or through incorporating the real-world costs our utility function omits (litigation risk, network effects, reputational damage; see Section **??**). The sensitivity analysis below quantifies the threshold at which the game transitions from exploitation to prisoner's dilemma.

The insurer's dominant defection is driven by a 116× per-transaction administrative cost asymmetry ($c_P = \$5.79$ vs. $c_I = \$0.05$). The insurer pays almost nothing for additional turns, so denial is nearly free. This formalizes a commonly described real-world dynamic: prior authorization imposes costs almost entirely on providers, giving insurers little incentive to streamline the process (American Medical Association, 2024).

Sensitivity to burden weight. The insurer's dominant defection is robust to the burden-weight parameters α_P and α_I. Because insurer admin costs are so low ($c_I = \$0.05$/turn), the insurer's strategy only changes when α_I reaches extreme values: the insurer becomes indifferent between cooperation and defection against a defecting provider at $\alpha_I \approx S^{DD}/\Delta C_I = \$2{,}274/(2 \times \$0.05) \approx 22{,}700$, meaning admin costs would need to be weighted ∼23,000× their dollar value to alter the insurer's incentive. Against a cooperating provider, the threshold is even higher ($\alpha_I \approx 28{,}300$). These thresholds drop dramatically if per-turn insurer costs rise to match provider costs (e.g., at $c_I = c_P = \$5.79$, the thresholds fall to $\alpha_I \approx 196$ and ≈ 244), values at which the game transitions to a prisoner's dilemma, consistent with the coordination-failure hypothesis of Moritz et al. (2025). Whether real-world insurer costs (incorporating litigation, network, and reputational factors) reach these levels is an empirical question our simulation does not resolve. The interactive sensitivity analysis in the supplementary materials allows exploration of all six parameters (R^{CC}, R^{DC}, S^{CD}, S^{DD}, α_P, α_I).

4.5. LLM Stochasticity

LLM outputs are stochastic: identical prompts and inputs can produce different outputs across runs. We observed this in two additional runs under the same strategy conditions.

Under CP_CI, the main run requested 2 lines using a bundled hepatitis panel (80074) and selected biosimilar Q5121 (Avsola), while an alternate run requested 4 lines with individual serologies (87340, 86704, 86706) and selected biosimilar Q5103 (Inflectra). Both reached full approval in 2

Table 4: Payoff matrix (U_P, U_I) at $\alpha_P = \alpha_I = 1$. Service values and admin costs in USD. Admin: $c_P = \$5.79$/turn, $c_I = \$0.05$/turn.

	Insurer Cooperate	**Insurer Defect**
Provider Cooperate	(\$1,868; −\$0.10)	(−\$23; \$2,828)
Provider Defect	(\$5,157; −\$0.10)	(−\$23; \$2,274)

turns, but through different service compositions and biosimilar selections.

Under CP_DI, stochastic variation produced qualitatively different outcomes. The main run lasted 4 turns: the provider resubmitted three times, eventually simplified to infliximab alone, and chose `ABANDON` with `NO_TREAT` (zero services delivered, provider payoff −\$23). An alternate run lasted only 2 turns: the provider resubmitted once with added hepatitis labs, then upon blanket denial chose `TREAT_ANYWAY` for the labs (subsequently paid at claims) but `NO_TREAT` for infliximab (partial delivery, provider payoff \$18). The same strategy pairing yielded payoffs of −\$23 versus \$18 depending solely on LLM sampling.

This stochasticity affects the qualitative game structure. With the main run's CP_DI payoff (−\$23), the provider is indifferent under insurer defection and defection is weakly dominant. With the alternate run's payoff (\$18), the provider strictly prefers cooperation under insurer defection, eliminating the provider's dominant strategy entirely and shifting the Nash equilibrium from (D,D) to (C,D). The game-theoretic classification is itself LLM-sampling-dependent, underscoring the need for multiple replications per condition.

5. Discussion

5.1. Gray Zone and Strategic Interpretation

With clinical facts and policy held constant across all nine conditions, strategy alone determined whether the patient received the drug, confirming that the case sits in a genuine interpretive gray zone. The key mechanism is the insurer's choice between `pending_info` and outright denial at initial review. Cooperating insurers used `pending_info` as a collaborative signal ("show me why this qualifies"), then accepted the cytopenias argument. Defecting and default insurers issued immediate denials and never used `pending_info` in any turn. This binary (pend versus deny) determined the drug coverage outcome across all nine conditions.

The default insurer's behavior (Section 4.3) suggests the `pending_info` mechanism is specifically tied to the cooperative strategy injection. Without explicit "prefer approve over deny" guidance, the LLM defaulted to literal policy application. This raises a question for real-world deployment: if AI adjudication systems are deployed without explicit interpretive guidance, our results suggest they may deny gray-zone treatments by default, not out of strategic cost-containment, but because literal policy application produces denial when criteria are ambiguous.

Strategy injection also governed provider behavior: service line volume (cooperate 2–5, defect 7–13, default 2–5), documentation aggressiveness, and willingness to appeal. Default providers behaved closer to cooperating providers in scope but diverged in composition, requesting imaging and IV administration codes that cooperating providers omitted.

Returning to the question posed in the introduction: does the coordination failure predicted by Moritz et al. (2025) emerge? At observed administrative costs, no. The game-theoretic analysis (Section 4.4) shows that the 116× cost asymmetry produces an exploitation structure, not a prisoner's dilemma: the insurer has a dominant strategy to defect because denial is nearly costless, and the provider's strategy is irrelevant under insurer defection. This is arguably worse than the predicted PD from the provider's perspective, because in a prisoner's dilemma both parties at least have an incentive to negotiate toward mutual cooperation, whereas in an exploitation game the insurer has no such incentive. The PD structure emerges only under richer cost models where insurer defection car-

ries meaningful consequences (litigation, network attrition, reputational damage), suggesting that the coordination-failure framing, while directionally correct, understates the severity of the current asymmetry.

6. Conclusion

We presented a multi-agent simulation framework where LLM-driven provider and insurer agents negotiate prior authorization for a specialty medication under information asymmetry. In a 3×3 factorial experiment on a single gray-zone clinical case, we found that the insurer's choice between requesting additional information and outright denial at initial review is the mechanism through which gray-zone ambiguity is resolved. Default (no-guidance) insurer agents denied the drug while approving diagnostics, applying policy literally rather than interpretively, suggesting that AI adjudication systems deployed without explicit guidance may default to denial in ambiguous cases. Game-theoretic analysis shows the game's classification is parameter-dependent: at observed administrative costs ($5.79/turn for providers vs. $0.05/turn for insurers, a 116× asymmetry), the game is an asymmetric exploitation where denial is nearly costless for insurers; under richer cost models incorporating litigation, reputational, and network costs, it can transition to a prisoner's dilemma. Extending this framework to multiple clinical cases, policy types, foundation models, and replicated runs would characterize the generality of these findings.

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Appendix A. Case Data

The patient case is represented as a structured JSON object with two top-level partitions. `patient_visible_data` contains all information available to both agents: demographics (age, sex), chief complaint, medical history, current medications, vital signs, and lab results (fecal calprotectin, albumin, leukocytes, platelets, serum calcium, serum magnesium, T-SPOT.TB). `environment_hidden_data` contains the ground-truth diagnosis, disease severity context, and the intended treatment request (infliximab 5 mg/kg); this partition is used by the Environment component for result synthesis but is not exposed to either strategic agent. The full case JSON is available in the supplementary materials.

Appendix B. Policy Documents

Each agent receives one document—the provider a clinical guideline, the insurer a coverage policy—creating the information asymmetry described in Section 3.1. Below we reproduce the coverage criteria as structured for agent consumption.

B.1. Provider: AGA 2021 Guidelines

The provider agent receives the following key recommendations:

1. Use anti-TNF therapy (e.g., infliximab) for induction and maintenance of remission in moderate to severe Crohn's disease.
2. In biologic-naïve moderate to severe Crohn's disease, infliximab is a recommended option for induction.
3. Prefer early introduction of a biologic (± immunomodulator) rather than delaying until after failure of mesalamine and/or corticosteroids.
4. Recommend against 5-ASA/mesalamine for induction or maintenance of remission in moderate to severe Crohn's disease.

The decision logic provided to the agent states: "If the clinical picture supports moderate-severe Crohn's disease (and/or high-risk/complicated features), biologic therapy such as infliximab is guideline-concordant even if the patient has not formally 'failed' mesalamine and/or corticosteroids."

B.2. Insurer: Cigna IP0660 Step-Therapy Criteria

The insurer agent receives Cigna Drug Coverage Policy IP0660 (effective 2026-01-01). For Crohn's disease initial therapy, the policy requires *all* of:

- Patient is $\geq$ 6 years of age.
- Patient meets **one** of the following (a, b, c, or d).
- The medication is prescribed by or in consultation with a gastroenterologist.

The step-therapy options are:

(a) Patient has tried or is currently taking corticosteroids, or corticosteroids are contraindicated. (Examples: prednisone, methylprednisolone.)

(b) Patient has tried one other conventional systemic therapy for Crohn's disease. (Examples: azathioprine, 6-mercaptopurine, methotrexate.)

(c) Patient has enterocutaneous (perianal or abdominal) or rectovaginal fistulas.

(d) Patient had ileocolonic resection (to reduce the chance of Crohn's disease recurrence).

Explicit exclusion: A trial of mesalamine does *not* count as a systemic therapy for Crohn's disease.
Exception: An exception to the requirement for a trial of or contraindication to steroids or a trial of one other conventional systemic agent can be made if the patient has already tried at

least one biologic other than the requested medication. A biosimilar of the requested biologic does not count.

The gray zone (Section 3.1) centers on option (a): the patient's cytopenias arguably make corticosteroids contraindicated, but no formal contraindication is documented in the case data. The full structured policy (including continuation criteria, dosing limits, and preferred product requirements) is available in the supplementary materials.

Appendix C. System Prompts

All agents use GPT-5 (Azure OpenAI) with JSON mode. Prompts follow a context engineering structure: the system prompt defines role, behavioral guidelines, strategy, and domain knowledge (stable across turns); the user prompt provides task instructions, context metadata, patient data, encounter history, and output schema (changes per turn). Below we reproduce the insurer system prompt template and all four strategy injection strings. The full prompt construction code (including provider prompts, action prompts, and Phase 3 prompts) is available in the supplementary materials.

C.1. Insurer System Prompt Template

The insurer agent receives the following system prompt (with {`strategy_block`} and {`policy_block`} injected per condition):

> You are an insurance utilization management team adjudicating a prior authorization request. Your goal: render coverage decisions based on policy criteria and clinical documentation.
>
> **ADMINISTRATIVE COST CONSIDERATION:** Each PA review costs ~$3.50 (manual) to ~$0.05 (electronic) in processing (CAQH 2023). Apply reasonableness standard: Do not request documentation already submitted. Do not pend repeatedly for the same item. If criteria cannot be met, deny clearly rather than pend indefinitely.
>
> {`strategy_block`}
>
> {`policy_block`}
>
> {`workflow_definitions`}

C.2. Strategy Injection Strings

Section 3.3 presents the core guidance for each strategy condition. The full strings injected into the system prompt include additional tactical framing (e.g., "Avoid adding extra line items or upgrading level-of-care unless clearly required" for provider cooperate; "demand stricter documentation and interpret ambiguity against approval" for insurer defect). These are injected under a `STRATEGY GUIDANCE` header. In the default condition, no strategy block is injected. The complete prompt construction code is available in the supplementary materials.

Appendix D. Agent Specifications and Action Spaces

This appendix provides the full technical specification of agent action spaces and request/response schemas summarized in Section **??**.

D.1. Service Line Types

Each provider request contains service lines classified as one of three types:

- **diagnostic_test**: Labs, imaging, pathology. When approved, the Environment returns results (ground-truth or synthesized).
- **treatment**: Medications and procedures. Subject to coverage policy criteria (e.g., step-therapy).
- **level_of_care**: Inpatient admission, observation, skilled nursing facility, home health. Determines the setting in which services are rendered.

D.2. Provider Action Space

After receiving an insurer response, the provider chooses one of two action modes:

RESUBMIT. Withdraw the entire authorization and start fresh at Level 0. All prior line statuses are reset. This is not a formal appeal; the insurer treats it as a new request. Use case: the provider's own submission contained errors (wrong procedure codes, missing documentation). Strategically, this avoids consuming appeal levels on fixable errors.

LINE_ACTIONS. Specify per-line responses for each non-approved line:

- `ACCEPT_MODIFY`: Accept the insurer's modification (for lines with status `modified`). The modified terms become the authorized service.
- `PROVIDE_DOCS`: Supply requested documentation (for lines with status `pending_info`). The insurer re-adjudicates at the same review level.
- `APPEAL`: Escalate to the next review level (for lines with status `denied` or `modified`). Use when the provider believes the submission was correct but the insurer misapplied policy.
- `ABANDON`: Give up on the line with one of two modes:
  - `NO_TREAT`: Patient does not receive the service.
  - `TREAT_ANYWAY`: Provider delivers the service and absorbs the cost (no insurer reimbursement).

The distinction between RESUBMIT and APPEAL is strategic: RESUBMIT corrects provider errors; APPEAL disputes insurer judgment.

Delivery assumption. Approved services are always delivered: if the insurer approves a line, the provider renders the service. Lines abandoned with `TREAT_ANYWAY` are also delivered, with the provider absorbing cost (no insurer reimbursement). All other abandoned lines are not delivered.

Phase 3 differences. Phase 3 (Claims Adjudication) processes billing for delivered services using the same turn structure as Phase 2. The provider may appeal or write off denied claims (`WRITE_OFF` replaces the Phase 2 abandon modes `NO_TREAT`/`TREAT_ANYWAY`).

D.3. Insurer Decision Space

Per service line, the insurer renders one of four decisions:

- `approved`: Authorization issued for the requested service.
- `modified`: Approved with changes to quantity, procedure code, or service parameters.
- `denied`: Does not meet coverage criteria.
- `pending_info`: Additional documentation requested. Not permitted at Level 2.

D.4. Request and Response Schemas

Request and response schemas derive from X12 278 (Health Care Services Review – prior authorization), X12 837 (Health Care Claim), and X12 835 (Claim Payment/Remittance Advice) transaction standards. Table 5 summarizes the key fields used in each phase. Full schema definitions and the complete X12 field mapping are available in the supplementary materials.

Appendix E. Interaction Logs

Each simulation run produces a structured audit log containing the complete sequence of provider submissions, insurer adjudications, provider actions, and Environment synthesis events, along with a metrics summary (turn counts, appeal counts, line outcomes). We develop an audit viewer tool for structured examination of these logs; it is available at `[link-placeholder]`. Raw audit logs for all nine conditions are included in the supplementary materials.

Appendix F. Review Level Design

Our three-level review structure (Section **??**) is grounded in the Medicare Fee-for-Service appeals process (Centers for Medicare & Medicaid Services, 2024). Medicare defines five levels of appeal that follow an initial coverage determination:

1. **Redetermination** by a Medicare Administrative Contractor (MAC)
2. **Reconsideration** by a Qualified Independent Contractor (QIC)
3. **Hearing** before an Administrative Law Judge (ALJ) within the Office of Medicare Hearings and Appeals
4. **Review** by the Medicare Appeals Council

Table 5: Key schema fields by phase, with X12 transaction set origin.

Phase	Direction	X12 Set	Key Fields
2	Provider → Insurer	278 Request	diagnosis_codes, procedure_code, code_type, request_type, requested_quantity, clinical_evidence
2	Insurer → Provider	278 Response	authorization_status, decision_reason, approved_quantity, authorization_number, modification_type, requested_documents
3	Provider → Insurer	837 Claim	procedure_code, authorization_number, clinical_documentation
3	Insurer → Provider	835 Remittance	adjudication_status, decision_reason, adjustment_group_code

5. **Judicial review** in federal district court

Critically, the initial coverage determination (in which the insurer first adjudicates a prior authorization request) is *not* itself an appeal level. The five-level appeal process begins only after a denial at initial determination. Table 6 maps our simulation levels to this structure.

We model through Level 2 because Levels 1–2 represent the administrative appeals process in which the dispute remains between the provider and insurer (or an independent reviewer). Levels 3–5 involve formal legal proceedings with different rules of evidence, representation requirements, and adjudicatory bodies, placing them outside the scope of an LLM-mediated negotiation simulation.

In practice, each real-world review level involves different reviewer types: initial determinations may be handled by utilization management nurses or automated systems, first-level appeals may involve medical director peer-to-peer review, and independent external reviews are conducted by entities with no relationship to the insurer. Our simulation does not encode these distinctions; the same insurer agent adjudicates at all levels (see Section **??** for discussion of this limitation).

Appendix G. Cost Data Sources

Service prices used for the payoff analysis (Section 4.4) are drawn from CMS-administered fee schedules. We assume a non-facility setting and a 60 kg patient receiving infliximab 5 mg/kg (300 mg = 30 billing units per infusion). Per-infusion drug prices = ASP per 10 mg × 30 units.

Drug prices (WV BMS ASP Pricing File, CMS national ASP + 6%, Q4 2025):

- J1745 Infliximab (Remicade): $932.70/infusion
- Q5121 Infliximab-axxq (Avsola): $612.15/infusion
- Q5104 Infliximab-dyyb (Renflexis): $809.97/infusion

The DP_CI insurer agent proposed "Q5105 Infliximab-abda (Renflexis)," but Q5105 is Retacrit (epoetin alfa-epbx). The correct HCPCS code for Renflexis is Q5104; we use Q5104 pricing.

Lab prices (WV BMS Clinical Lab Fee Schedule, 04/2025–03/2026):

- 80074 Acute Hepatitis Panel: $42.87
- 80053 Comprehensive Metabolic Panel: $9.50
- 85025 CBC with Differential: $6.99
- 86140 C-Reactive Protein: $4.66
- 87340 HBsAg: $9.30
- 86704 Anti-HBc (total): $10.85
- 86706 Anti-HBs: $9.67
- 86803 HCV Antibody: $12.84
- 83993 Fecal Calprotectin: $17.67

Infusion administration and imaging (benchmark estimates, bounded below by verified CMS PFS facility rates, calibrated against CY 2025 CF = $32.3465):

Table 6: Mapping of simulation review levels to Medicare FFS appeal structure.

Simulation	Medicare Equivalent	Reviewer
Level 0	Initial Determination	Insurer agent
Level 1	Redetermination (MAC)	Insurer agent
Level 2	Reconsideration (QIC/IRE)	Insurer agent
Not modeled	ALJ Hearing	–
Not modeled	Appeals Council	–
Not modeled	Federal Court	–

- 96413 Chemo IV infusion, initial hr: ∼$130
- 96365 Therapeutic IV infusion, initial hr: ∼$70
- 74183 MRI abdomen w/wo contrast: ∼$350
- 72197 MRI pelvis w/wo contrast: ∼$350

These four codes collectively account for <15% of total service value in any condition; varying by ±50% does not alter the game-theoretic classification.

Administrative costs (2023 CAQH Index, electronic prior authorization): $c_P = \$5.79$/turn (provider), $c_I = \$0.05$/turn (insurer).

Appendix H. Phase Architecture

Phase 1: Patient Presentation. Phase 1 currently loads the patient case deterministically from a published case report. The architecture is designed to support probabilistic noise injection: introducing missing labs, transcription errors, or incomplete medication histories into the patient record before agents see it. The primary motivation is ecological validity: patient data errors are among the most common real-world denial triggers, and modeling them is necessary to capture this class of PA friction. A secondary benefit is that noise injection generates multiple realistic case variations from a single published artifact, addressing the practical scarcity of publicly available PA cases with complete clinical data.

Phase 4: Financial Settlement. Phase 4 currently computes provider reimbursement and insurer payment per service line based on CMS fee schedules (Appendix G). The architecture is designed to also calculate patient out-of-pocket costs (copays, coinsurance, deductible contributions) and, in future work, to feed settlement outcomes back into Phase 1 as a repeated game. This feedback loop would enable modeling of market-level dynamics that emerge over repeated interactions: patient plan-switching driven by denial experiences, provider network attrition, insurer algorithm adaptation in response to detected documentation patterns, and reputation effects. These dynamics matter because the pathologies of interest (market failure, trust erosion, AI arms race escalation) emerge from repeated games, not single episodes. The current single-episode design captures individual outcomes; the Phase 4-to-Phase 1 feedback loop is the architectural hook for this extension.

Supplementary Materials. The full codebase, including case data JSON, complete policy files, prompt construction code, raw interaction logs, and X12 schema field mappings, is available at `[link-placeholder]`.